IASP

Inventory & Supply Chain Analytics Pipeline

End-to-End Data Engineering Project Report

Prepared by	Rachit Jain
Role	Data Engineering Intern
Organization	Celebal Technologies
Architecture	Medallion Architecture — Bronze → Silver → Gold
Core Tools	Microsoft Azure, Databricks, PySpark, Delta Lake, Power BI

Purpose: This report documents the complete implementation in beginner-friendly language, showing how six raw supply-chain datasets were stored, processed, validated, transformed into business-ready tables, and visualized in Power BI.

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1. Executive Summary

IASP is an end-to-end inventory and supply-chain analytics pipeline built as a practical data engineering project. The project starts with six CSV datasets, stores the raw source data in Microsoft Azure, processes the data in Databricks with PySpark and Delta tables, and follows the Medallion Architecture: **Bronze** → **Silver** → **Gold**.

The Bronze layer preserves source data, the Silver layer cleans and validates it, and the Gold layer creates business-facing analytical tables. The final outputs are used to build a Power BI dashboard containing revenue, units sold, inventory value, low-stock alerts, delivery performance, product performance, and warehouse-level analysis.

The implementation is intentionally appropriate for an internship-level project. It demonstrates core concepts learned in data engineering—cloud storage, Spark transformations, Delta tables, data quality, referential integrity, Slowly Changing Dimension Type 2, aggregations, and dashboarding—without adding unnecessary advanced AI or production infrastructure.

2. Business Problem and Objectives

Inventory and supply-chain data is usually distributed across products, stock, suppliers, warehouses, transactions, and shipments. Raw files may contain duplicate records, missing values, inconsistent text, invalid references, or historical versions. Directly connecting such data to a dashboard can produce unreliable results.

Project objective: create a structured pipeline that converts raw operational data into trusted, analytics-ready outputs.

Business Question	Project Output
What is the current inventory value?	inventory_snapshot
Which items require replenishment attention?	low_stock_alert
Which products generate the most revenue?	sales_summary
How does revenue change month by month?	sales_trend
How is stock moving?	product_movement
How are suppliers performing on delivery?	supplier_performance

3. Technology Stack

Technology	Use in the Project
Microsoft Azure Storage	Raw landing area for the six source datasets
Azure Databricks Free Edition	Notebook-based data engineering environment
Apache Spark / PySpark	Data reading, cleaning, joins, validation and aggregation
Delta Lake	Storage format for Bronze, Silver and Gold tables
Unity Catalog schemas	Logical organization of workspace.bronze, workspace.silver and workspace.gold
Power BI Desktop	Interactive business dashboard
GitHub	Project documentation and submission repository

4. Source Datasets

Dataset	Raw Records	Purpose
products	2,000	Product master and historical product versions
inventory	10,000	Stock by product and warehouse
suppliers	20	Supplier master information
warehouses	10	Warehouse master information
transactions	20,000	Sales / inventory transactions
shipments	5,000	Shipment and delivery information
Total	37,030	Combined Bronze ingestion volume

5. End-to-End Architecture

The project follows a layered data flow. Each layer has one clear responsibility, making the pipeline easier to understand, test, and maintain.

RAW SOURCE	→	BRONZE	→	SILVER	→	GOLD	→	POWER BI
6 CSV files		Raw Delta tables		Clean + validated		Business KPIs		Dashboard

Raw: source files are retained as the landing input. **Bronze:** source data is ingested with minimal transformation. **Silver:** data is cleaned, standardized and validated. **Gold:** trusted data is joined and aggregated for business reporting. **Power BI:** Gold outputs are presented as KPIs and interactive visuals.

6. Raw Data Storage in Microsoft Azure

The Azure storage container contains a **raw** area with separate folders for inventory, products, shipments, suppliers, transactions, and warehouses. This provides a clear source landing structure before transformation begins.

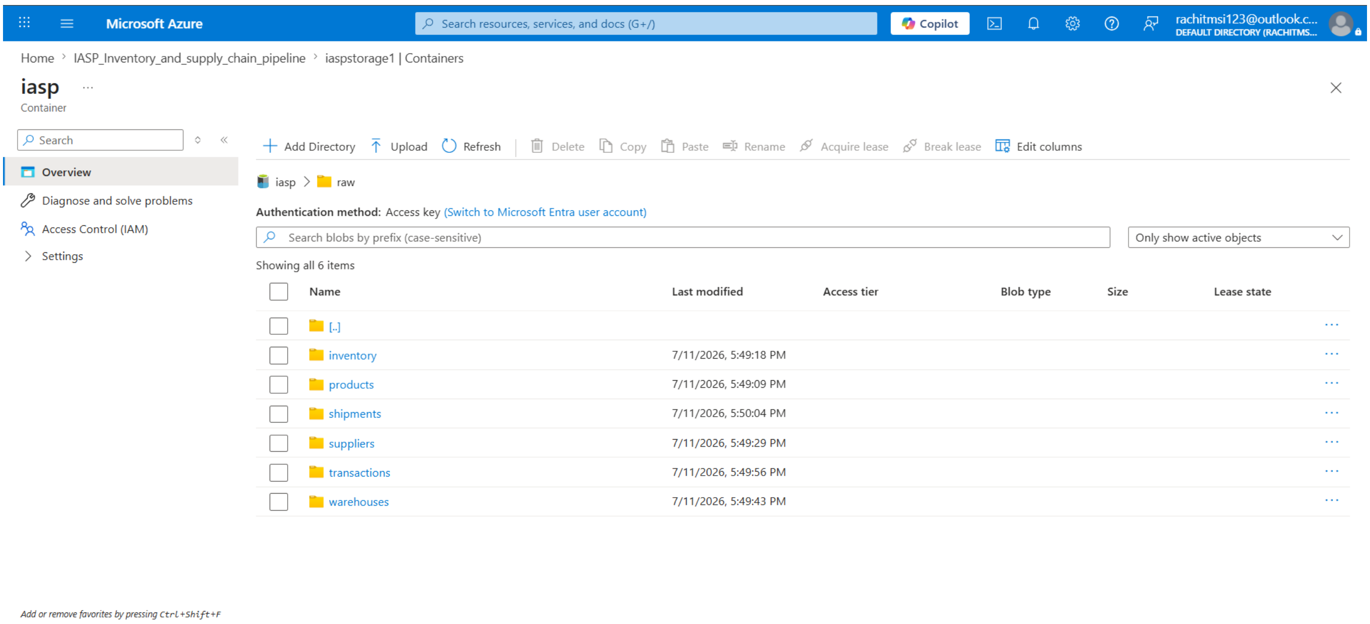

Figure 1. Azure raw storage structure containing the six source dataset folders.

This screenshot is important evidence that the project begins with cloud-hosted raw data rather than directly creating dashboard tables.

7. Bronze Layer — Raw Ingestion

7.1 Purpose

The Bronze layer is the first processing layer. Its purpose is to preserve the source data in a structured Delta-table form with minimal business transformation. At this stage, the focus is ingestion and traceability—not heavy cleaning.

7.2 Databricks Workspace

The implementation is organized into separate notebooks for project setup, Bronze ingestion, Silver transformations and validation, and Gold analytics. This separation keeps the pipeline modular and easy to rerun.

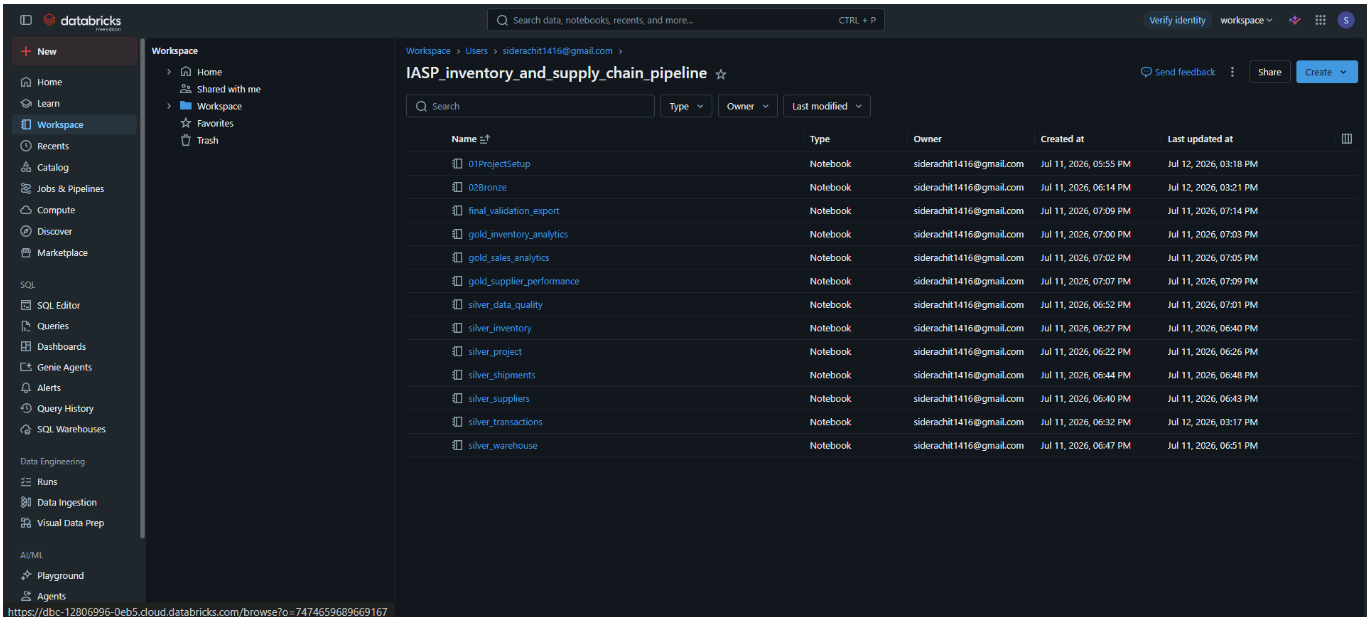

Figure 2. Databricks workspace showing the project notebooks.

7.3 Bronze Ingestion Result

All six datasets were loaded into the Bronze schema. The ingestion summary confirms a total of **37,030 raw records**.

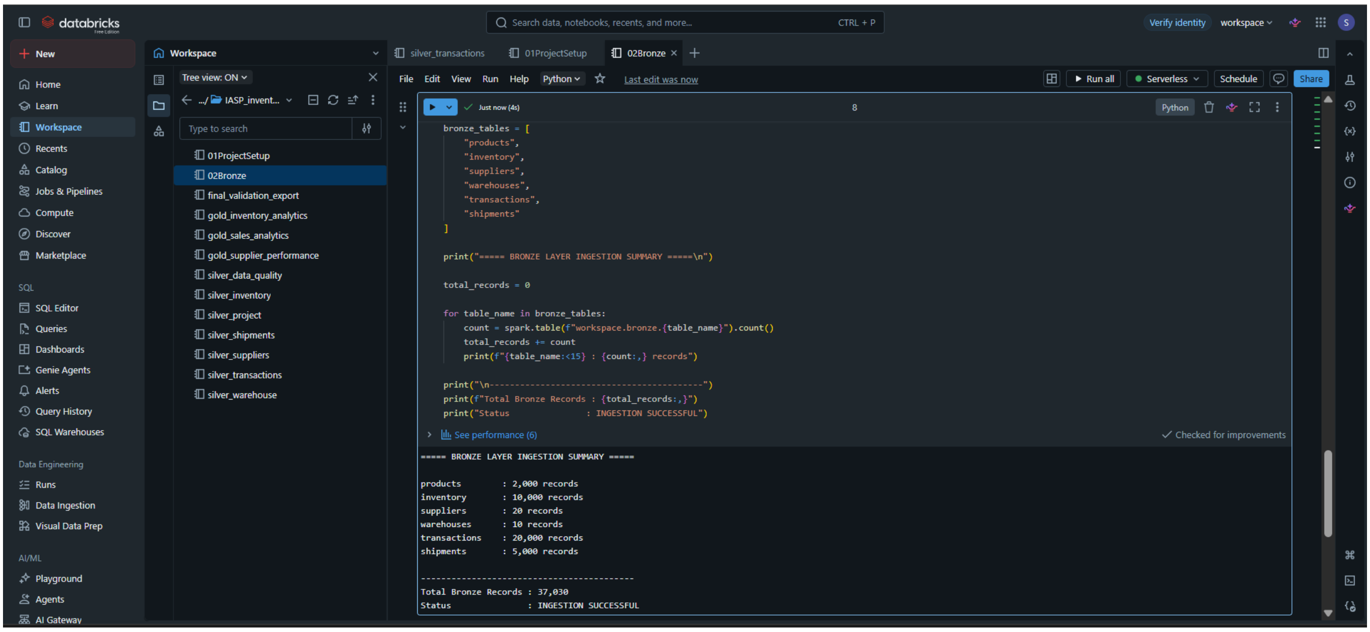

Figure 3. Bronze layer ingestion summary and record counts.

7.4 Example of Bronze Data

The Bronze products table still resembles the source data. Values such as mixed-case category paths and null brand values remain visible because Bronze is intended to preserve the raw source representation.

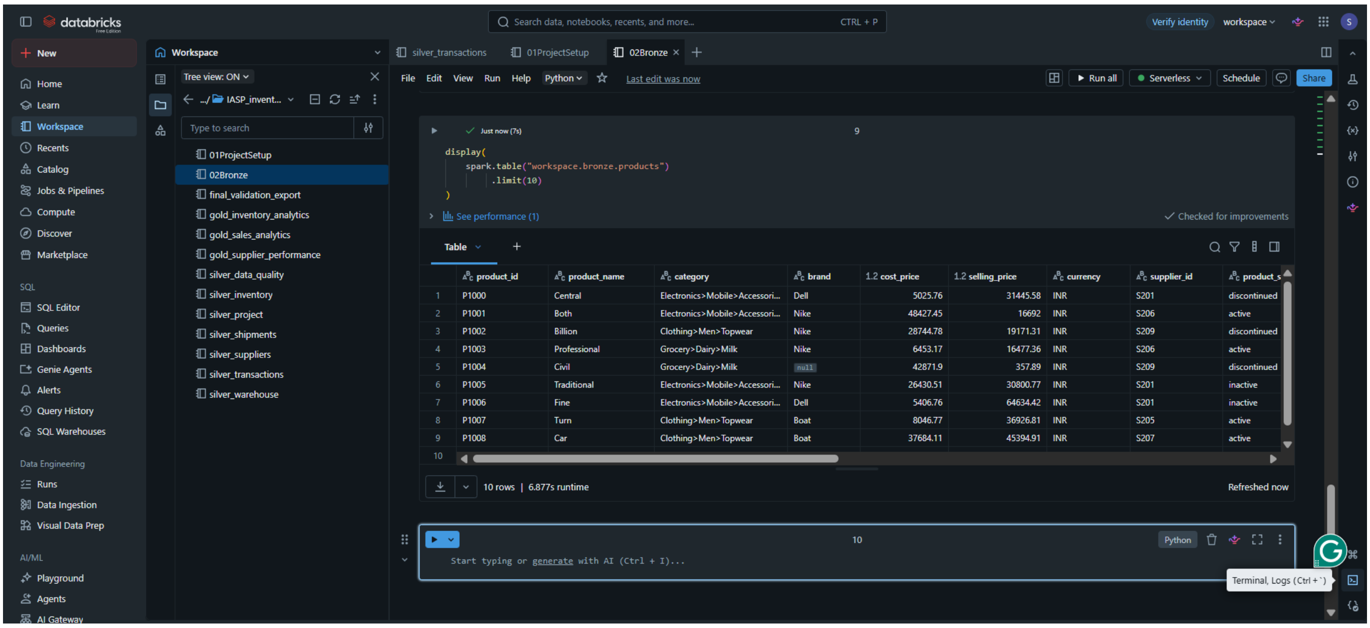

Figure 4. Example records from workspace.bronze.products.

Key idea: Bronze answers, “What exactly did we ingest?” Silver answers, “What data can we trust and use?”

8. Silver Layer — Cleaning and Validation

8.1 Purpose

The Silver layer converts raw records into trusted datasets. Typical operations include type casting, text standardization, duplicate handling, null checks, business-rule validation, date conversion, and referential-integrity checks.

Dataset	Main Silver Responsibility
products	Standardization plus SCD Type 2 history management
inventory	Validate product/warehouse references and usable stock records
transactions	Validate product references and transaction fields
suppliers	Clean and validate supplier master data
shipments	Validate supplier references and delivery information
warehouses	Clean and validate warehouse master data

8.2 Silver Validation Results

After cleaning and validation, the trusted current datasets contain 500 current products, 4,821 inventory records, 15,894 transaction records, 14 supplier records, 5,000 shipment records, and 10 warehouse records.

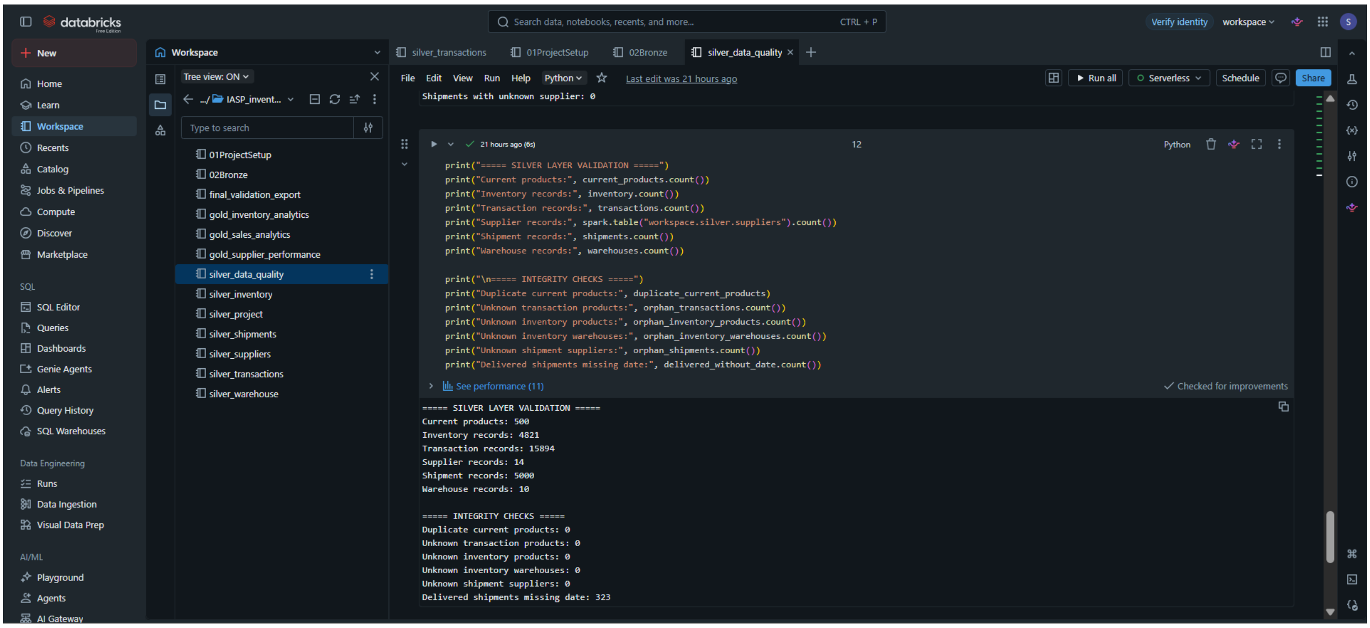

Figure 5. Silver-layer validation and integrity-check output.

The integrity checks show zero duplicate current products and zero unknown references for transaction products, inventory products, inventory warehouses, and shipment suppliers. The validation also identifies **323 delivered shipments with missing delivery dates**. This is kept visible as a data-quality finding rather than hidden.

8.3 Example of Cleaned and Historical Product Data

The Silver products table includes standardized fields and history-management columns such as start_date, end_date, and current_flag.

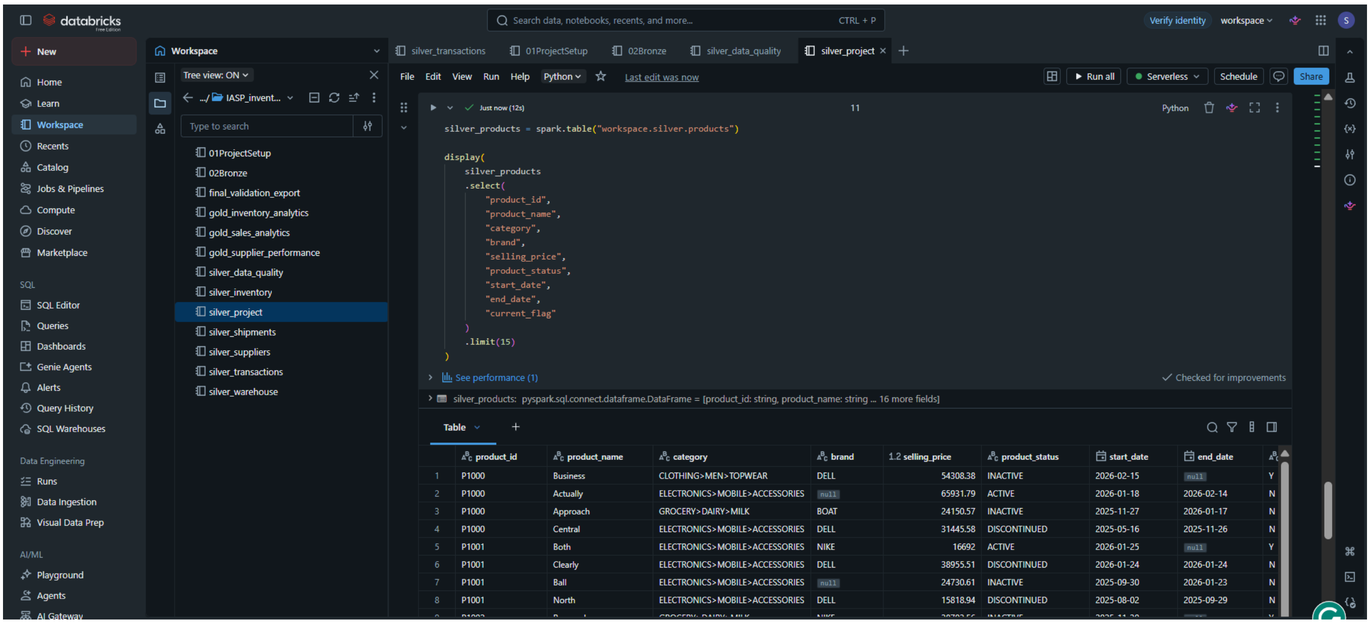

Figure 6. Cleaned Silver product data with historical version columns.

9. Slowly Changing Dimension Type 2 (SCD2)

SCD Type 2 is used when the project must preserve previous versions of a business entity instead of overwriting them. In this project, a product can change over time—for example, its name, category, brand, selling price, or status.

Column	Meaning
product_id	Business key used to identify the product

Column	Meaning
start_date	Date from which a version became effective
end_date	Date on which the version stopped being effective
current_flag	Y for the current version and N for a historical version

For product **P1000**, four versions are visible. The first three have an end date and current_flag = N. The latest version has no end date and current_flag = Y. Therefore, history is preserved while the current record remains easy to identify.

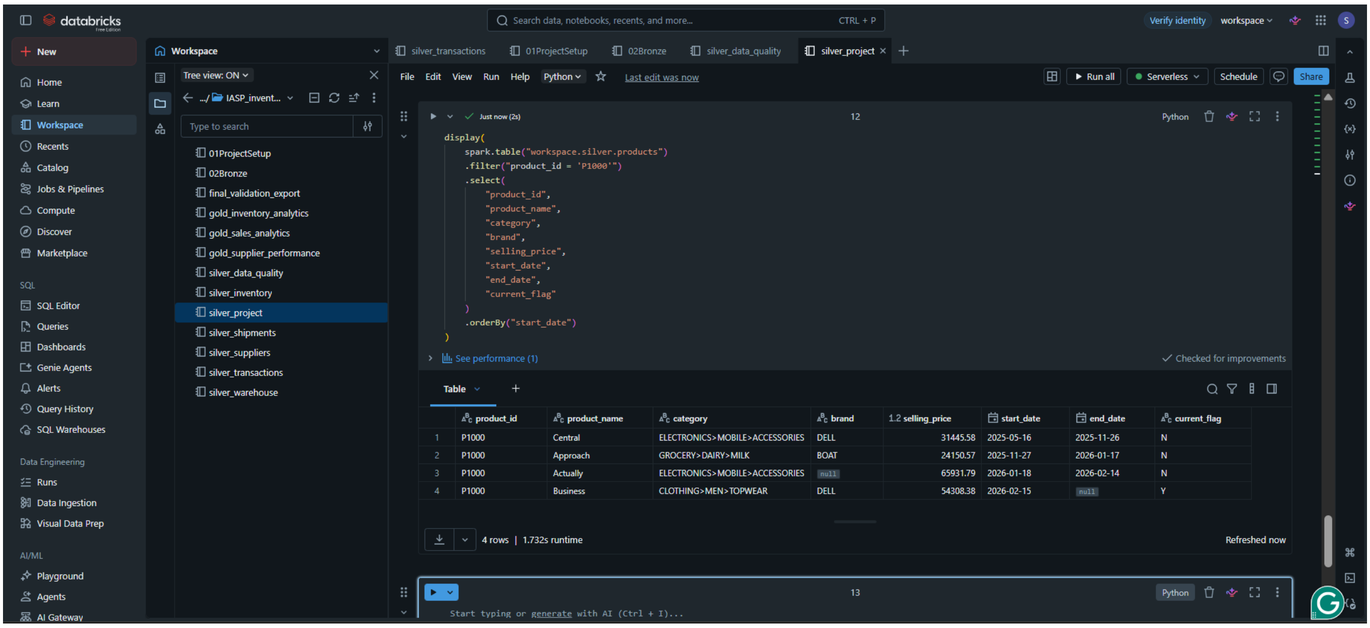

Figure 7. SCD Type 2 example for product P1000.

Simple example: instead of changing an old product row permanently, the old row is closed with an end date and a new current row is retained. This allows historical reporting.

10. Data Quality and Referential Integrity

Data quality checks prevent invalid records from silently entering the analytical layer. The project checks duplicate current product versions and verifies that IDs used in transaction, inventory and shipment data exist in their corresponding master datasets.

Check	Observed Result
Duplicate current products	0
Unknown transaction products	0
Unknown inventory products	0
Unknown inventory warehouses	0
Unknown shipment suppliers	0
Delivered shipments missing delivery date	323 — retained as a visible quality issue

The presence of a known quality issue does not mean the validation failed. A good pipeline should expose quality problems clearly so that users understand the limitations of the source data.

11. Gold Layer — Business Analytics

11.1 Purpose

The Gold layer transforms trusted Silver data into tables designed around business questions. These tables are easier for reporting tools to consume because the required joins, calculations and aggregations have already been performed.

11.2 Gold Tables Created

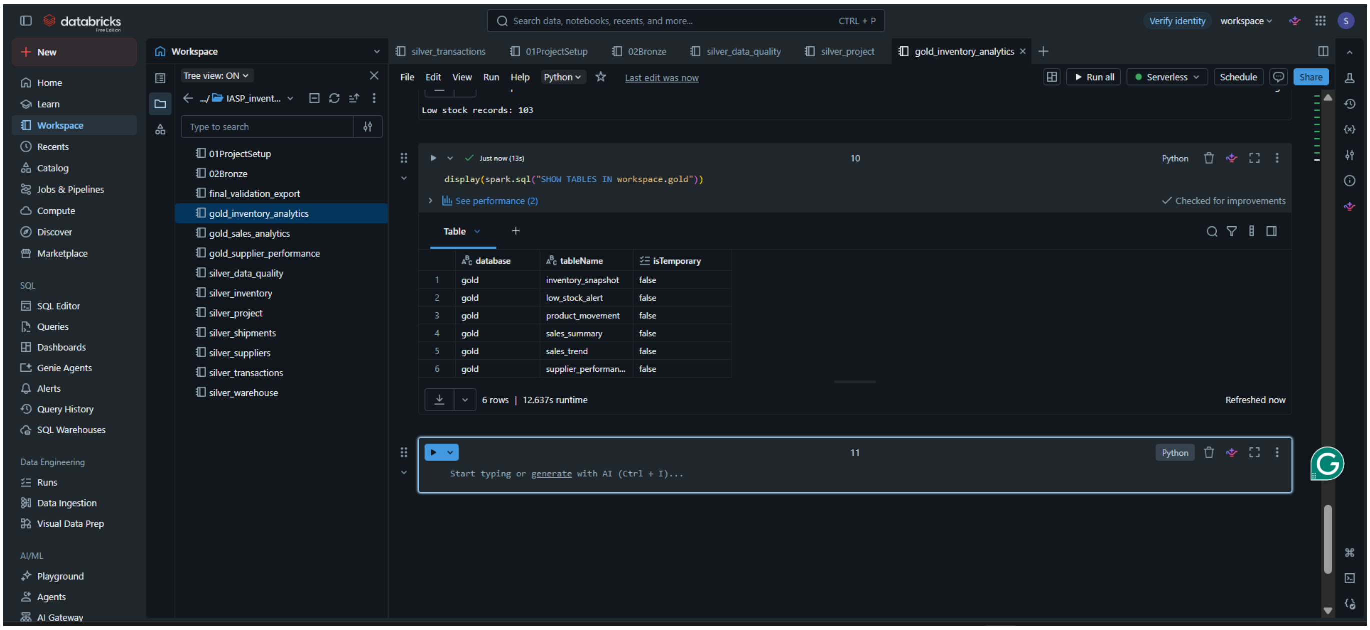

Figure 8. Six business-facing tables in workspace.gold.

Gold Table	Purpose
inventory_snapshot	Current stock position and inventory value
low_stock_alert	Products requiring replenishment attention
product_movement	Product-level stock movement analysis
sales_summary	Revenue and units sold by product
sales_trend	Monthly revenue trend
supplier_performance	Supplier delivery performance

11.3 Example: Low-Stock Business Output

The low-stock output combines warehouse, product, available stock, reorder level, and an alert level. This converts operational stock values into an action-oriented business view.

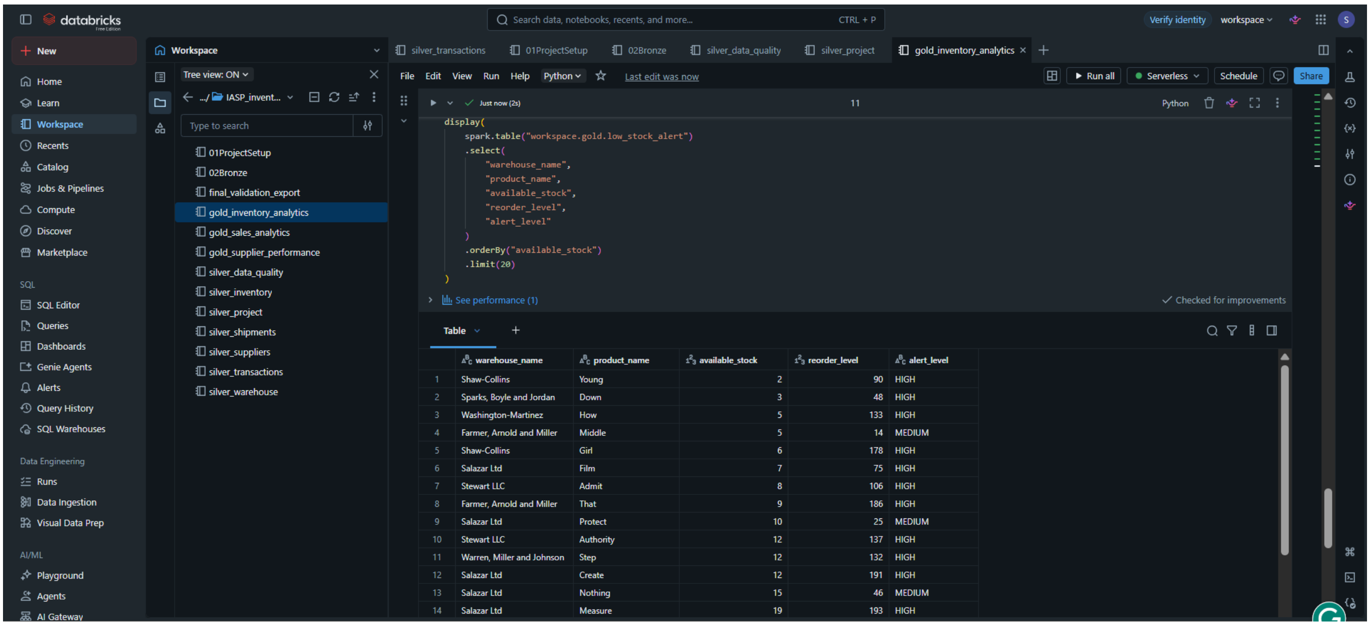

Figure 9. Example Gold low-stock alert output.

For example, if available stock is substantially below the reorder level, the record is highlighted with a higher alert category. The final Gold table therefore answers a practical question: *Which products need replenishment attention?*

12. Power BI Dashboard

The final dashboard presents the Gold-layer outputs in a single executive view. It contains KPI cards, trend charts, product analysis, supplier performance, warehouse inventory value, low-stock alerts, and slicers.

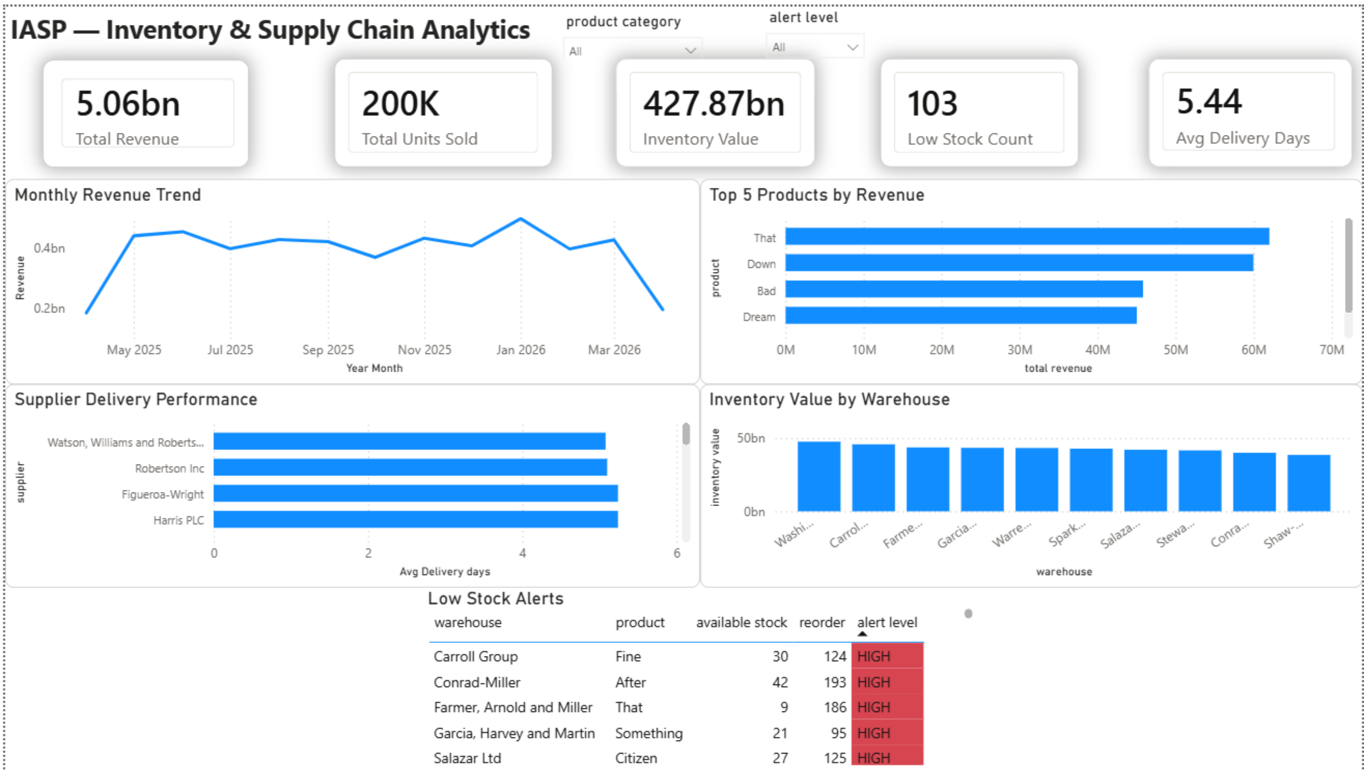

Figure 10. Final IASP Power BI executive dashboard.

KPI	Displayed Value
Total Revenue	5.06bn

KPI	Displayed Value
Total Units Sold	200K
Inventory Value	427.87bn
Low Stock Count	103
Average Delivery Days	5.44

The dashboard also includes a monthly revenue trend, top products by revenue, supplier delivery performance, inventory value by warehouse, and a detailed low-stock alert table. Product-category and alert-level slicers allow interactive filtering.

13. End-to-End Example: How One Business Insight Is Produced

Consider the question: **Which products are at risk of running low on stock?**

Step	What Happens
1. Raw	Inventory, product and warehouse source files arrive in the raw storage area.
2. Bronze	The source records are ingested into Bronze Delta tables with minimal transformation.
3. Silver	Invalid references and unusable records are checked; product and warehouse data is standardized.
4. Gold	Trusted data is joined and available stock is compared with reorder information to create low_stock_alert.
5. Power BI	The dashboard shows the low-stock count and a detailed alert table for business users.

This example demonstrates why the layers are useful: the dashboard does not need to repeat raw-data cleaning logic. It consumes a business-ready Gold output.

14. Challenges and Practical Adaptations

Challenge	Practical Resolution
Free-edition platform limitations	Used the capabilities available in the Databricks Free Edition environment and organized tables through
Source schema differences	Transformations were aligned with the actual source columns instead of forcing assumptions.
Historical product versions	Implemented SCD Type 2 using start_date, end_date and current_flag.
Invalid or incomplete source data	Added validation and referential-integrity checks before Gold processing.
Missing shipment delivery dates	Reported the issue explicitly as a data-quality finding.

The report describes the implementation that was actually completed. It does not claim real-time streaming, CI/CD, advanced AI, or other production features that were not part of the internship implementation.

15. Requirement-to-Implementation Mapping

The project follows the core intent of the provided inventory and supply-chain pipeline specification while adapting the implementation to the available free-tier environment.

Expected Concept	Implemented Evidence
Cloud raw storage	Azure raw folders for all six datasets

Expected Concept	Implemented Evidence
Bronze layer	workspace.bronze Delta tables and ingestion summary
Silver layer	Cleaning, validation, current product logic and integrity checks
SCD Type 2	Historical product versions with start/end dates and current flag
Gold layer	Six business-facing analytical tables
Business KPIs	Revenue, units, inventory value, low stock and delivery performance
Visualization	Interactive Power BI executive dashboard

16. Results, Limitations and Future Scope

16.1 Results

The project successfully demonstrates a complete data journey from raw cloud storage to business visualization. It processes 37,030 Bronze records, validates the Silver layer, preserves product history, creates six Gold analytical tables, and surfaces the outputs through an executive Power BI dashboard.

16.2 Current Limitations

The project is a batch-oriented internship implementation. It does not include continuous streaming, enterprise orchestration, automated CI/CD, formal monitoring infrastructure, or a full rejected-record quarantine framework.

16.3 Future Improvements

Possible future improvements include scheduled Databricks workflows, incremental ingestion, quarantine tables for rejected records, automated data-quality testing, monitoring and alerting, governed BI connectivity, and CI/CD. These are future extensions rather than features claimed in the current implementation.

17. Conclusion

IASP demonstrates the fundamental workflow of a modern data engineering project in a clear and practical way. Raw supply-chain data is stored in Azure, ingested into a Bronze layer, cleaned and validated in a Silver layer, converted into business-ready Gold tables, and visualized in Power BI.

The project applies core concepts such as Medallion Architecture, PySpark transformations, Delta tables, data quality checks, referential integrity, SCD Type 2, business aggregation, and dashboarding. The implementation remains suitable for an intern-level project while still showing a complete end-to-end understanding of how raw data becomes useful business information.

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